

In [10]:

```
myDict = {"Java":1, "C++":2, "Python":3, "JavaScript":2, 'Perl':4}
print(myDict)
```

```
{'Java': 1, 'C++': 2, 'Python': 3, 'JavaScript': 2, 'Perl': 4}
```

In [16]:

```
for value in myDict.values():
    print(value, end = " ")
print(' ')

setofvalue = set({})
for value in myDict.values():
    setofvalue.add(value)
print(setofvalue)
```

```
1 2 3 2 4
{1, 2, 3, 4}
```

In [17]:

```
uniquevalues = list({val for val in myDict.values() })
print(uniquevalues)
```

```
[1, 2, 3, 4]
```

In [19]:

```
#print sum of key-value pairs in dictionary
myDict1 = {2:4, 3:9, 4:16 , 5:25}
# output = [6,12,20,30]
sumlist = []
for key in myDict1:
    sumlist.append(key + myDict1[key])
print(sumlist)
```

```
[6, 12, 20, 30]
```

In [20]:

```
myDict1[5] = 125
print(myDict1)
```

```
{2: 4, 3: 9, 4: 16, 5: 125}
```

In [21]:

```
myDict2 = {2:4, 5:25, 10:20, 4:16, 3:9, 1:2}
print(myDict2)
```

```
{2: 4, 5: 25, 10: 20, 4: 16, 3: 9, 1: 2}
```

In [31]:

```
#sort the key:value pair in the dict (myDict2) based on keys
print(myDict2.keys())
keylist2 = list(myDict2.keys())
sortedkeylist2 = sorted(keylist2)
print(sortedkeylist2)
```

```
dict_keys([2, 5, 10, 4, 3, 1])
[1, 2, 3, 4, 5, 10]
```

In [33]:

```
myDict3 = {}
for key in sortedkeylist2:
    myDict3[key] = myDict2[key]    #myDict3[key] = value
print(myDict3)
```

```
{1: 2, 2: 4, 3: 9, 4: 16, 5: 25, 10: 20}
```

In [2]:

```
record1 = {'id': 101, 'name': 'Shivang Yadav', 'Age': 21}
record2 = {'id': 101, 'Age': 21, 'name': 'Shivang Yadav'}
record3 = {'id': 102, 'name': 'Radib Kar', 'Age': 23}

if record1 == record2:
    print("record 1 and record 2 are same")
else:
    print("record 1 and record 2 are not same")
if record1 == record3:
    print("record 1 and record 3 are same")
else:
    print("record 1 and record 3 are not same")
```

```
record 1 and record 2 are same
record 1 and record 3 are not same
```

In [40]:

```
pip install deepdiff
```

Collecting deepdiff

Downloading deepdiff-5.8.1-py3-none-any.whl (69 kB)

Collecting ordered-set<4.2.0,>=4.1.0

Downloading ordered_set-4.1.0-py3-none-any.whl (7.6 kB)

Installing collected packages: ordered-set, deepdiff

Successfully installed deepdiff-5.8.1 ordered-set-4.1.0

Note: you may need to restart the kernel to use updated packages.

In [6]:

```

from deepdiff import DeepDiff

diff12 = DeepDiff(record1, record2)
print(diff12)

print("=====")
diff13 = DeepDiff(record1, record3)
print(diff13)

{}
=====
{'values_changed': {"root['id']": {'new_value': 102, 'old_value': 101}, "root['name']": {'new_value': 'Radib Kar', 'old_value': 'Shivang Yadav'}, "root['Age']": {'new_value': 23, 'old_value': 21}}}

```

In [7]:

```

#CSV Files
import csv
with open('G:\\Python Session\\python.csv') as csv_file:
    csv_reader = csv.reader(csv_file, delimiter = ',')
    for row in csv_reader:
        print(row)

```

```

['name', 'department', 'Age']
['Sahil', 'Accounting', '45']
['Joe', 'IT', '54']
['Rohan', 'CSE', '34']
['Sachit', 'HR', '23']

```

In [8]:

```

import pandas
df = pandas.read_csv('G:\\Python Session\\python.csv')
print(df)

```

	name	department	Age
0	Sahil	Accounting	45
1	Joe	IT	54
2	Rohan	CSE	34
3	Sachit	HR	23

In [9]:

```

import csv
filename = 'G:\\Python Session\\python.csv'
with open(filename, 'r') as data:
    for line in csv.DictReader(data):
        print(line)

```

```

OrderedDict([('name', 'Sahil'), ('department', 'Accounting'), ('Age', '45')])
OrderedDict([('name', 'Joe'), ('department', 'IT'), ('Age', '54')])
OrderedDict([('name', 'Rohan'), ('department', 'CSE'), ('Age', '34')])
OrderedDict([('name', 'Sachit'), ('department', 'HR'), ('Age', '23')])

```

In [29]:

```
import csv
filename = 'G:\\Python Session\\python.csv'
with open(filename, 'r') as csv_file:
    csv_reader = csv.DictReader(csv_file)
    count_line = 0
    for row in csv_file:
        print(f'{" ".join(row)}')
        count_line += 1
```

name,department,Age

Sahil,Accounting,45

Joe,IT,54

Rohan,CSE,34

Sachit,HR,23

In [30]:

```
data = [{'Rank': 'B', 'first_name': 'Parker', 'last_name': 'Brian'},
{'Rank': 'A', 'first_name': 'Smith', 'last_name': 'Rodriguez'},
{'Rank': 'C', 'first_name': 'Tom', 'last_name': 'smith'},
{'Rank': 'B', 'first_name': 'Jane', 'last_name': 'Oscar'},
{'Rank': 'A', 'first_name': 'Alex', 'last_name': 'Tim'}]
print(data)
```

```
[{'Rank': 'B', 'first_name': 'Parker', 'last_name': 'Brian'}, {'Rank': 'A', 'first_name': 'Smith', 'last_name': 'Rodriguez'}, {'Rank': 'C', 'first_name': 'Tom', 'last_name': 'smith'}, {'Rank': 'B', 'first_name': 'Jane', 'last_name': 'Oscar'}, {'Rank': 'A', 'first_name': 'Alex', 'last_name': 'Tim'}]
```

In [31]:

```
#write dictionary into the csv file
import csv
with open('G:\\Python Session\\data.csv', 'w') as csv_file:
    columnnames = ["Rank", "first_name", "last_name"]
    writer = csv.DictWriter(csv_file, fieldnames=columnnames)
    writer.writeheader()

    writer.writerow({'Rank': 'B', 'first_name': 'Parker', 'last_name': 'Brian'})
    writer.writerow({'Rank': 'A', 'first_name': 'Smith', 'last_name': 'Rodriguez'})
    writer.writerow({'Rank': 'B', 'first_name': 'Jane', 'last_name': 'Oscar'})
    writer.writerow({'Rank': 'B', 'first_name': 'Jane', 'last_name': 'Loive'})

print("Done")
```

Done

In [39]:

```
def readfile(filename):  
    import csv  
    with open(filename, 'r') as csv_file:  
        csv_reader = csv.DictReader(csv_file)  
        count_line = 0  
        for row in csv_file:  
            print(f'{" ".join(row)}', end = "\t")  
            count_line += 1
```

In [40]:

```
readfile('G:\\Python Session\\data.csv')
```

Rank,first_name,last_name

B,Parker,Brian

A,Smith,Rodriguez

B,Jane,Oscar

B,Jane,Loive

In [36]:

```
#write dictionary into the csv file  
import csv  
with open('G:\\Python Session\\data1.csv','w') as csv_file:  
    columnnames = ["Rank","first_name","last_name"]  
    writer = csv.DictWriter(csv_file, fieldnames=columnnames)  
    writer.writeheader()  
  
    # data --> dictionary  
    for var in data:  
        writer.writerow(var)  
  
print("Done")
```

Done

In [43]:

```
print(data)
print('=====')
```

readfile('G:\\Python Session\\data1.csv')

```
[{'Rank': 'B', 'first_name': 'Parker', 'last_name': 'Brian'}, {'Rank': 'A', 'first_name': 'Smith', 'last_name': 'Rodriguez'}, {'Rank': 'C', 'first_name': 'Tom', 'last_name': 'smith'}, {'Rank': 'B', 'first_name': 'Jane', 'last_name': 'Oscar'}, {'Rank': 'A', 'first_name': 'Alex', 'last_name': 'Tim'}]
```

```
=====
```

```
Rank,first_name,last_name
```

```
B,Parker,Brian
```

```
A,Smith,Rodriguez
```

```
C,Tom,smith
```

```
B,Jane,Oscar
```

```
A,Alex,Tim
```

In [44]:

```
readfile('G:\\Python Session\\archive\\StudentsPerformance.csv')
```

gender,race/ethnicity,parental level of education,lunch,test preparation course,math score,reading score,writing score

female,group B,bachelor's degree,standard,none,72,72,74
 female,group C,some college,standard,completed,69,90,88
 female,group B,master's degree,standard,none,90,95,93
 male,group A,associate's degree,free/reduced,none,47,57,44
 male,group C,some college,standard,none,76,78,75
 female,group B,associate's degree,standard,none,71,83,78
 female,group B,some college,standard,completed,88,95,92
 male,group B,some college,free/reduced,none,40,43,39
 male,group D,high school,free/reduced,completed,64,64,67
 female,group B,high school,free/reduced,none,38,60,50
 male,group C,associate's degree,standard,none,58,54,52
 male,group D,associate's degree,standard,none,40,52,43
 female,group B,high school,standard,none,65,81,73
 male,group A,some college,standard,completed,78,72,70
 female,group A,master's degree,standard,none,50,53,58
 female,group C,some high school,standard,none,69,75,78
 male,group C,high school,standard,none,88,89,86
 female,group B,some high school,free/reduced,none,18,32,28
 male,group C,master's degree,free/reduced,completed,46,42,46
 female,group C,associate's degree,free/reduced,none,54,58,61
 male,group D,high school,standard,none,66,69,63
 female,group B,some college,free/reduced,completed,65,75,70
 male,group D,some college,standard,none,44,54,53
 female,group C,some high school,standard,none,69,73,73
 male,group D,bachelor's degree,free/reduced,completed,74,71,80
 male,group A,master's degree,free/reduced,none,73,74,72
 male,group B,some college,standard,none,69,54,55
 female,group C,bachelor's degree,standard,none,67,69,75
 male,group C,high school,standard,none,70,70,65
 female,group D,master's degree,standard,none,62,70,75
 female,group D,some college,standard,none,69,74,74
 female,group B,some college,standard,none,63,65,61
 female,group E,master's degree,free/reduced,none,56,72,65
 male,group D,some college,standard,none,40,42,38
 male,group E,some college,standard,none,97,87,82
 male,group E,associate's degree,standard,completed,81,81,79
 female,group D,associate's degree,standard,none,74,81,83
 female,group D,some high school,free/reduced,none,50,64,59
 female,group D,associate's degree,free/reduced,completed,75,90,88
 male,group B,associate's degree,free/reduced,none,57,56,57
 male,group C,associate's degree,free/reduced,none,55,61,54
 female,group C,associate's degree,standard,none,58,73,68
 female,group B,associate's degree,standard,none,53,58,65
 male,group B,some college,free/reduced,completed,59,65,66
 female,group E,associate's degree,free/reduced,none,50,56,54
 male,group B,associate's degree,standard,none,65,54,57
 female,group A,associate's degree,standard,completed,55,65,62
 female,group C,high school,standard,none,66,71,76
 female,group D,associate's degree,free/reduced,completed,57,74,76
 male,group C,high school,standard,completed,82,84,82
 male,group E,some college,standard,none,53,55,48
 male,group E,associate's degree,free/reduced,completed,77,69,68
 male,group C,some college,standard,none,53,44,42
 male,group D,high school,standard,none,88,78,75
 female,group C,some high school,free/reduced,completed,71,84,87
 female,group C,high school,free/reduced,none,33,41,43
 female,group E,associate's degree,standard,completed,82,85,86
 male,group D,associate's degree,standard,none,52,55,49
 male,group D,some college,standard,completed,58,59,58

female,group C,some high school,free/reduced,none,0,17,10
male,group E,bachelor's degree,free/reduced,completed,79,74,72
male,group A,some high school,free/reduced,none,39,39,34
male,group A,associate's degree,free/reduced,none,62,61,55
female,group C,associate's degree,standard,none,69,80,71
female,group D,some high school,standard,none,59,58,59
male,group B,some high school,standard,none,67,64,61
male,group D,some high school,free/reduced,none,45,37,37
female,group C,some college,standard,none,60,72,74
male,group B,associate's degree,free/reduced,none,61,58,56
female,group C,associate's degree,standard,none,39,64,57
female,group D,some college,free/reduced,completed,58,63,73
male,group D,some college,standard,completed,63,55,63
female,group A,associate's degree,free/reduced,none,41,51,48
male,group C,some high school,free/reduced,none,61,57,56
male,group C,some high school,standard,none,49,49,41
male,group B,associate's degree,free/reduced,none,44,41,38
male,group E,some high school,standard,none,30,26,22
male,group A,bachelor's degree,standard,completed,80,78,81
female,group D,some high school,standard,completed,61,74,72
female,group E,master's degree,standard,none,62,68,68
female,group B,associate's degree,standard,none,47,49,50
male,group B,high school,free/reduced,none,49,45,45
male,group A,some college,free/reduced,completed,50,47,54
male,group E,associate's degree,standard,none,72,64,63
male,group D,high school,free/reduced,none,42,39,34
female,group C,some college,standard,none,73,80,82
female,group C,some college,free/reduced,none,76,83,88
female,group D,associate's degree,standard,none,71,71,74
female,group A,some college,standard,none,58,70,67
female,group D,some high school,standard,none,73,86,82
female,group C,bachelor's degree,standard,none,65,72,74
male,group C,high school,free/reduced,none,27,34,36
male,group C,high school,standard,none,71,79,71
male,group C,associate's degree,free/reduced,completed,43,45,50
female,group B,some college,standard,none,79,86,92
male,group C,associate's degree,free/reduced,completed,78,81,82
male,group B,some high school,standard,completed,65,66,62
female,group E,some college,standard,completed,63,72,70
female,group D,some college,free/reduced,none,58,67,62
female,group D,bachelor's degree,standard,none,65,67,62
male,group B,some college,standard,none,79,67,67
male,group D,bachelor's degree,standard,completed,68,74,74
female,group D,associate's degree,standard,none,85,91,89
male,group B,high school,standard,completed,60,44,47
male,group C,some college,standard,completed,98,86,90
female,group C,some college,standard,none,58,67,72
female,group D,master's degree,standard,none,87,100,100
male,group E,associate's degree,standard,completed,66,63,64
female,group B,associate's degree,free/reduced,none,52,76,70
female,group B,some high school,standard,none,70,64,72
female,group D,associate's degree,free/reduced,completed,77,89,98
male,group C,high school,standard,none,62,55,49
male,group A,associate's degree,standard,none,54,53,47
female,group D,some college,standard,none,51,58,54
female,group E,bachelor's degree,standard,completed,99,100,100
male,group C,high school,standard,none,84,77,74
female,group B,bachelor's degree,free/reduced,none,75,85,82
female,group D,bachelor's degree,standard,none,78,82,79
female,group D,some high school,standard,none,51,63,61
female,group C,some college,standard,none,55,69,65

female,group C,bachelor's degree,standard,completed,79,92,89
male,group B,associate's degree,standard,completed,91,89,92
female,group C,some college,standard,completed,88,93,93
male,group D,high school,free/reduced,none,63,57,56
male,group E,some college,standard,none,83,80,73
female,group B,high school,standard,none,87,95,86
male,group B,some high school,standard,none,72,68,67
male,group D,some college,standard,completed,65,77,74
male,group D,master's degree,standard,none,82,82,74
female,group A,bachelor's degree,standard,none,51,49,51
male,group D,master's degree,standard,none,89,84,82
male,group C,some high school,free/reduced,completed,53,37,40
male,group E,some college,free/reduced,completed,87,74,70
female,group C,some college,standard,completed,75,81,84
male,group D,bachelor's degree,free/reduced,completed,74,79,75
male,group C,bachelor's degree,standard,none,58,55,48
male,group B,some high school,standard,completed,51,54,41
male,group E,high school,standard,none,70,55,56
female,group C,associate's degree,standard,none,59,66,67
male,group D,some college,standard,completed,71,61,69
female,group D,some high school,standard,none,76,72,71
female,group C,some college,free/reduced,none,59,62,64
female,group E,some college,free/reduced,completed,42,55,54
male,group A,high school,standard,none,57,43,47
male,group D,some college,standard,none,88,73,78
female,group C,some college,free/reduced,none,22,39,33
male,group B,some high school,standard,none,88,84,75
male,group C,associate's degree,free/reduced,none,73,68,66
female,group D,bachelor's degree,standard,completed,68,75,81
male,group E,associate's degree,free/reduced,completed,100,100,93
male,group A,some high school,standard,completed,62,67,69
male,group A,bachelor's degree,standard,none,77,67,68
female,group B,associate's degree,standard,completed,59,70,66
male,group D,bachelor's degree,standard,none,54,49,47
male,group D,some high school,standard,none,62,67,61
female,group C,some college,standard,completed,70,89,88
female,group E,high school,free/reduced,completed,66,74,78
male,group B,some college,free/reduced,none,60,60,60
female,group B,associate's degree,standard,completed,61,86,87
male,group D,associate's degree,free/reduced,none,66,62,64
male,group B,associate's degree,free/reduced,completed,82,78,74
female,group E,some college,free/reduced,completed,75,88,85
male,group B,master's degree,free/reduced,none,49,53,52
male,group C,high school,standard,none,52,53,49
female,group E,master's degree,standard,none,81,92,91
female,group C,bachelor's degree,standard,completed,96,100,100
male,group C,high school,free/reduced,completed,53,51,51
female,group B,master's degree,free/reduced,completed,58,76,78
female,group B,high school,standard,completed,68,83,78
female,group C,some college,free/reduced,completed,67,75,70
male,group A,high school,standard,completed,72,73,74
male,group E,some high school,standard,none,94,88,78
female,group D,some college,standard,none,79,86,81
female,group C,associate's degree,standard,none,63,67,70
female,group C,bachelor's degree,free/reduced,completed,43,51,54
female,group C,master's degree,standard,completed,81,91,87
female,group B,high school,free/reduced,completed,46,54,58
female,group C,associate's degree,standard,completed,71,77,77
female,group B,master's degree,free/reduced,completed,52,70,62
female,group D,some high school,standard,completed,97,100,100
male,group C,master's degree,free/reduced,completed,62,68,75

female,group C,some college,free/reduced,none,46,64,66
female,group E,high school,standard,none,50,50,47
female,group D,associate's degree,standard,none,65,69,70
male,group C,some high school,free/reduced,completed,45,52,49
male,group C,associate's degree,free/reduced,completed,65,67,65
male,group E,high school,standard,none,80,76,65
male,group D,some high school,standard,completed,62,66,68
male,group B,some high school,free/reduced,none,48,52,45
female,group C,bachelor's degree,standard,none,77,88,87
female,group E,associate's degree,standard,none,66,65,69
male,group D,some college,standard,completed,76,83,79
female,group B,some high school,standard,none,62,64,66
male,group D,some college,standard,completed,77,62,62
female,group C,master's degree,standard,completed,69,84,85
male,group D,associate's degree,standard,none,61,55,52
male,group C,some high school,free/reduced,completed,59,69,65
male,group E,high school,free/reduced,none,55,56,51
female,group B,some college,free/reduced,none,45,53,55
female,group B,bachelor's degree,free/reduced,none,78,79,76
female,group C,associate's degree,standard,completed,67,84,86
female,group D,some college,free/reduced,none,65,81,77
male,group C,associate's degree,standard,none,69,77,69
female,group B,associate's degree,standard,none,57,69,68
male,group C,some college,standard,none,59,41,42
male,group D,some high school,standard,completed,74,71,78
male,group E,bachelor's degree,standard,none,82,62,62
male,group E,high school,standard,completed,81,80,76
female,group B,some college,free/reduced,none,74,81,76
female,group B,some college,free/reduced,none,58,61,66
male,group D,some high school,free/reduced,completed,80,79,79
male,group C,some college,free/reduced,none,35,28,27
female,group C,high school,free/reduced,none,42,62,60
male,group C,associate's degree,free/reduced,completed,60,51,56
male,group E,high school,standard,completed,87,91,81
male,group B,some high school,standard,completed,84,83,75
female,group E,associate's degree,free/reduced,completed,83,86,88
female,group C,high school,free/reduced,none,34,42,39
male,group B,high school,free/reduced,none,66,77,70
male,group B,some high school,standard,completed,61,56,56
female,group D,high school,standard,completed,56,68,74
male,group B,associate's degree,standard,none,87,85,73
female,group C,some high school,free/reduced,none,55,65,62
male,group D,some high school,standard,none,86,80,75
female,group B,associate's degree,standard,completed,52,66,73
female,group E,master's degree,free/reduced,none,45,56,54
female,group C,some college,standard,none,72,72,71
male,group D,high school,standard,none,57,50,54
male,group A,some high school,free/reduced,none,68,72,64
female,group C,some college,standard,completed,88,95,94
male,group D,some college,standard,none,76,64,66
male,group C,associate's degree,standard,none,46,43,42
female,group B,bachelor's degree,standard,none,67,86,83
male,group E,some high school,standard,none,92,87,78
male,group C,bachelor's degree,standard,completed,83,82,84
male,group D,associate's degree,standard,none,80,75,77
male,group D,bachelor's degree,free/reduced,none,63,66,67
female,group D,some high school,standard,completed,64,60,74
male,group B,some college,standard,none,54,52,51
male,group C,associate's degree,standard,none,84,80,80
male,group D,high school,free/reduced,completed,73,68,66
female,group E,bachelor's degree,standard,none,80,83,83

female,group D,high school,standard,none,56,52,55
male,group E,some college,standard,none,59,51,43
male,group D,some high school,standard,none,75,74,69
male,group C,associate's degree,standard,none,85,76,71
male,group E,associate's degree,standard,none,89,76,74
female,group B,high school,standard,completed,58,70,68
female,group B,high school,standard,none,65,64,62
male,group C,high school,standard,none,68,60,53
male,group A,some high school,standard,completed,47,49,49
female,group D,some college,free/reduced,none,71,83,83
female,group B,some high school,standard,completed,60,70,70
male,group D,master's degree,standard,none,80,80,72
male,group D,high school,standard,none,54,52,52
female,group E,some college,standard,none,62,73,70
female,group C,associate's degree,free/reduced,none,64,73,68
male,group C,associate's degree,standard,completed,78,77,77
female,group B,some college,standard,none,70,75,78
female,group C,master's degree,free/reduced,completed,65,81,81
female,group C,some high school,free/reduced,completed,64,79,77
male,group C,some college,standard,completed,79,79,78
female,group C,some high school,free/reduced,none,44,50,51
female,group E,high school,standard,none,99,93,90
male,group D,high school,standard,none,76,73,68
male,group D,some high school,free/reduced,none,59,42,41
female,group C,bachelor's degree,standard,none,63,75,81
female,group D,high school,standard,none,69,72,77
female,group D,associate's degree,standard,completed,88,92,95
female,group E,some college,free/reduced,none,71,76,70
male,group C,bachelor's degree,standard,none,69,63,61
male,group C,some college,standard,none,58,49,42
female,group D,associate's degree,free/reduced,none,47,53,58
female,group D,some college,standard,none,65,70,71
male,group B,some college,standard,completed,88,85,76
male,group C,bachelor's degree,standard,none,83,78,73
female,group C,some high school,standard,completed,85,92,93
female,group E,high school,standard,completed,59,63,75
female,group C,some high school,free/reduced,none,65,86,80
male,group B,bachelor's degree,free/reduced,none,73,56,57
male,group D,high school,standard,none,53,52,42
male,group D,high school,standard,none,45,48,46
female,group D,bachelor's degree,free/reduced,none,73,79,84
female,group D,some college,free/reduced,completed,70,78,78
female,group B,some high school,standard,none,37,46,46
male,group B,associate's degree,standard,completed,81,82,82
male,group E,associate's degree,standard,completed,97,82,88
female,group B,some high school,standard,none,67,89,82
male,group B,bachelor's degree,free/reduced,none,88,75,76
male,group E,some high school,standard,completed,77,76,77
male,group C,associate's degree,standard,none,76,70,68
male,group D,some high school,standard,none,86,73,70
male,group C,some high school,standard,completed,63,60,57
female,group E,bachelor's degree,standard,none,65,73,75
male,group D,high school,free/reduced,completed,78,77,80
male,group B,associate's degree,free/reduced,none,67,62,60
male,group A,some high school,standard,completed,46,41,43
male,group E,associate's degree,standard,completed,71,74,68
male,group C,high school,free/reduced,completed,40,46,50
male,group D,associate's degree,free/reduced,none,90,87,75
male,group A,some college,free/reduced,completed,81,78,81
male,group D,some high school,free/reduced,none,56,54,52
female,group C,associate's degree,standard,completed,67,84,81

male,group B,associate's degree,standard,none,80,76,64
female,group C,associate's degree,standard,completed,74,75,83
male,group A,some college,standard,none,69,67,69
male,group E,some college,standard,completed,99,87,81
male,group C,some high school,standard,none,51,52,44
female,group B,associate's degree,free/reduced,none,53,71,67
female,group D,high school,free/reduced,none,49,57,52
female,group B,associate's degree,standard,none,73,76,80
male,group B,bachelor's degree,standard,none,66,60,57
male,group D,bachelor's degree,standard,completed,67,61,68
female,group C,associate's degree,free/reduced,completed,68,67,69
female,group C,bachelor's degree,standard,completed,59,64,75
male,group C,high school,standard,none,71,66,65
female,group D,master's degree,standard,completed,77,82,91
male,group C,associate's degree,standard,none,83,72,78
male,group B,bachelor's degree,standard,none,63,71,69
female,group D,associate's degree,free/reduced,none,56,65,63
female,group C,high school,free/reduced,completed,67,79,84
female,group E,high school,standard,none,75,86,79
female,group C,some college,standard,none,71,81,80
female,group C,some high school,free/reduced,none,43,53,53
female,group C,high school,free/reduced,none,41,46,43
female,group C,some college,standard,none,82,90,94
male,group C,some college,standard,none,61,61,62
male,group A,some college,free/reduced,none,28,23,19
male,group C,associate's degree,standard,completed,82,75,77
female,group B,some high school,standard,none,41,55,51
male,group C,high school,standard,none,71,60,61
male,group C,associate's degree,standard,none,47,37,35
male,group E,associate's degree,standard,completed,62,56,53
male,group B,associate's degree,standard,none,90,78,81
female,group C,bachelor's degree,standard,none,83,93,95
female,group B,some college,free/reduced,none,61,68,66
male,group D,some high school,standard,completed,76,70,69
male,group C,associate's degree,standard,none,49,51,43
female,group B,some high school,free/reduced,none,24,38,27
female,group D,some high school,free/reduced,completed,35,55,60
male,group C,high school,free/reduced,none,58,61,52
female,group C,high school,standard,none,61,73,63
female,group B,high school,standard,completed,69,76,74
male,group D,associate's degree,standard,completed,67,72,67
male,group D,some college,standard,none,79,73,67
female,group C,high school,standard,none,72,80,75
male,group B,some college,standard,none,62,61,57
female,group C,bachelor's degree,standard,completed,77,94,95
male,group D,high school,free/reduced,none,75,74,66
male,group E,associate's degree,standard,none,87,74,76
female,group B,bachelor's degree,standard,none,52,65,69
male,group E,some college,standard,none,66,57,52
female,group C,some college,standard,completed,63,78,80
female,group C,associate's degree,standard,none,46,58,57
female,group C,some college,standard,none,59,71,70
female,group B,bachelor's degree,standard,none,61,72,70
male,group A,associate's degree,standard,none,63,61,61
female,group C,some college,free/reduced,completed,42,66,69
male,group D,some college,free/reduced,none,59,62,61
female,group D,some college,standard,none,80,90,89
female,group B,high school,standard,none,58,62,59
male,group B,some high school,standard,completed,85,84,78
female,group C,some college,standard,none,52,58,58
female,group D,some high school,free/reduced,none,27,34,32

male,group C,some college,standard,none,59,60,58
male,group A,bachelor's degree,free/reduced,completed,49,58,60
male,group C,high school,standard,completed,69,58,53
male,group C,bachelor's degree,free/reduced,none,61,66,61
female,group A,some high school,free/reduced,none,44,64,58
female,group D,some high school,standard,none,73,84,85
male,group E,some college,standard,none,84,77,71
female,group C,some college,free/reduced,completed,45,73,70
male,group D,some high school,standard,none,74,74,72
female,group D,some college,standard,completed,82,97,96
female,group D,bachelor's degree,standard,none,59,70,73
male,group E,associate's degree,free/reduced,none,46,43,41
female,group D,some high school,standard,none,80,90,82
female,group D,master's degree,free/reduced,completed,85,95,100
female,group A,some high school,standard,none,71,83,77
male,group A,bachelor's degree,standard,none,66,64,62
female,group B,associate's degree,standard,none,80,86,83
male,group C,associate's degree,standard,completed,87,100,95
male,group C,master's degree,free/reduced,none,79,81,71
female,group E,some high school,free/reduced,none,38,49,45
female,group A,some high school,free/reduced,none,38,43,43
female,group E,some college,standard,none,67,76,75
female,group E,bachelor's degree,standard,none,64,73,70
female,group C,associate's degree,free/reduced,none,57,78,67
female,group D,high school,standard,none,62,64,64
male,group D,master's degree,standard,none,73,70,75
male,group E,some high school,free/reduced,completed,73,67,59
female,group D,some college,standard,none,77,68,77
male,group E,some college,standard,none,76,67,67
male,group C,associate's degree,standard,completed,57,54,56
female,group C,some high school,standard,completed,65,74,77
male,group A,high school,free/reduced,none,48,45,41
female,group B,high school,free/reduced,none,50,67,63
female,group C,associate's degree,standard,none,85,89,95
male,group B,some high school,standard,none,74,63,57
male,group D,some high school,standard,none,60,59,54
female,group C,some high school,standard,completed,59,54,67
male,group A,some college,standard,none,53,43,43
female,group A,some college,free/reduced,none,49,65,55
female,group D,high school,standard,completed,88,99,100
female,group C,high school,standard,none,54,59,62
female,group C,some high school,standard,none,63,73,68
male,group B,associate's degree,standard,completed,65,65,63
female,group B,associate's degree,standard,none,82,80,77
female,group D,high school,free/reduced,completed,52,57,56
male,group D,associate's degree,standard,completed,87,84,85
female,group D,master's degree,standard,completed,70,71,74
male,group E,some college,standard,completed,84,83,78
male,group D,associate's degree,standard,none,71,66,60
male,group B,some high school,standard,completed,63,67,67
female,group C,bachelor's degree,free/reduced,completed,51,72,79
male,group E,high school,standard,none,84,73,69
male,group C,bachelor's degree,standard,completed,71,74,68
male,group C,associate's degree,standard,none,74,73,67
male,group D,some college,standard,none,68,59,62
male,group E,high school,free/reduced,completed,57,56,54
female,group C,associate's degree,free/reduced,completed,82,93,93
female,group D,high school,standard,completed,57,58,64
female,group D,master's degree,free/reduced,completed,47,58,67
female,group A,some high school,standard,completed,59,85,80
male,group B,some college,free/reduced,none,41,39,34

female,group C,some college,free/reduced,none,62,67,62
male,group C,bachelor's degree,standard,none,86,83,86
male,group C,some high school,free/reduced,none,69,71,65
male,group A,some high school,free/reduced,none,65,59,53
male,group C,some high school,free/reduced,none,68,63,54
male,group C,associate's degree,free/reduced,none,64,66,59
female,group C,high school,standard,none,61,72,70
male,group C,high school,standard,none,61,56,55
female,group A,some high school,free/reduced,none,47,59,50
male,group C,some high school,standard,none,73,66,66
male,group C,some college,free/reduced,completed,50,48,53
male,group D,associate's degree,standard,none,75,68,64
male,group D,associate's degree,free/reduced,none,75,66,73
male,group C,high school,standard,none,70,56,51
male,group D,some high school,standard,completed,89,88,82
female,group C,some college,standard,completed,67,81,79
female,group D,high school,standard,none,78,81,80
female,group A,some high school,free/reduced,none,59,73,69
female,group B,associate's degree,standard,none,73,83,76
male,group A,some high school,free/reduced,none,79,82,73
female,group C,some high school,standard,completed,67,74,77
male,group D,some college,free/reduced,none,69,66,60
male,group C,high school,standard,completed,86,81,80
male,group B,high school,standard,none,47,46,42
male,group B,associate's degree,standard,none,81,73,72
female,group C,some college,free/reduced,completed,64,85,85
female,group E,some college,standard,none,100,92,97
female,group C,associate's degree,free/reduced,none,65,77,74
male,group C,some college,free/reduced,none,65,58,49
female,group C,associate's degree,free/reduced,none,53,61,62
male,group C,bachelor's degree,free/reduced,none,37,56,47
female,group D,bachelor's degree,standard,none,79,89,89
male,group D,associate's degree,free/reduced,none,53,54,48
female,group E,bachelor's degree,standard,none,100,100,100
male,group B,high school,standard,completed,72,65,68
male,group C,bachelor's degree,free/reduced,none,53,58,55
male,group B,some college,free/reduced,none,54,54,45
female,group E,some college,standard,none,71,70,76
female,group C,some college,free/reduced,none,77,90,91
male,group A,bachelor's degree,standard,completed,75,58,62
female,group C,some college,standard,none,84,87,91
female,group D,associate's degree,free/reduced,none,26,31,38
male,group A,high school,free/reduced,completed,72,67,65
female,group A,high school,free/reduced,completed,77,88,85
male,group C,some college,standard,none,91,74,76
female,group C,associate's degree,standard,completed,83,85,90
female,group C,high school,standard,none,63,69,74
female,group C,associate's degree,standard,completed,68,86,84
female,group D,some high school,standard,none,59,67,61
female,group B,associate's degree,standard,completed,90,90,91
female,group D,bachelor's degree,standard,completed,71,76,83
male,group E,bachelor's degree,standard,completed,76,62,66
male,group D,associate's degree,standard,none,80,68,72
female,group D,master's degree,standard,none,55,64,70
male,group E,associate's degree,standard,none,76,71,67
male,group B,high school,standard,completed,73,71,68
female,group D,associate's degree,free/reduced,none,52,59,56
male,group C,some college,free/reduced,none,68,68,61
male,group A,high school,standard,none,59,52,46
female,group B,associate's degree,standard,none,49,52,54
male,group C,high school,standard,none,70,74,71

male,group D,some college,free/reduced,none,61,47,56
female,group C,associate's degree,free/reduced,none,60,75,74
male,group B,some high school,standard,completed,64,53,57
male,group A,associate's degree,free/reduced,completed,79,82,82
female,group A,associate's degree,free/reduced,none,65,85,76
female,group C,associate's degree,standard,none,64,64,70
female,group C,some college,standard,none,83,83,90
female,group C,bachelor's degree,standard,none,81,88,90
female,group B,high school,standard,none,54,64,68
male,group D,high school,standard,completed,68,64,66
female,group C,some college,standard,none,54,48,52
female,group D,some college,free/reduced,completed,59,78,76
female,group B,some high school,standard,none,66,69,68
male,group E,some college,standard,none,76,71,72
female,group D,master's degree,standard,none,74,79,82
female,group B,associate's degree,standard,completed,94,87,92
male,group C,some college,free/reduced,none,63,61,54
female,group E,associate's degree,standard,completed,95,89,92
female,group D,master's degree,free/reduced,none,40,59,54
female,group B,some high school,standard,none,82,82,80
male,group A,high school,standard,none,68,70,66
male,group B,bachelor's degree,free/reduced,none,55,59,54
male,group C,master's degree,standard,none,79,78,77
female,group C,bachelor's degree,standard,none,86,92,87
male,group D,some college,standard,none,76,71,73
male,group A,some high school,standard,none,64,50,43
male,group D,some high school,free/reduced,none,62,49,52
female,group B,some high school,standard,completed,54,61,62
female,group B,master's degree,free/reduced,completed,77,97,94
female,group C,some high school,standard,completed,76,87,85
female,group D,some college,standard,none,74,89,84
female,group E,some college,standard,completed,66,74,73
female,group D,some high school,standard,completed,66,78,78
female,group B,high school,free/reduced,completed,67,78,79
male,group D,some college,standard,none,71,49,52
female,group C,associate's degree,standard,none,91,86,84
male,group D,bachelor's degree,standard,none,69,58,57
male,group C,master's degree,free/reduced,none,54,59,50
male,group C,high school,standard,completed,53,52,49
male,group E,some college,standard,none,68,60,59
male,group C,some high school,free/reduced,completed,56,61,60
female,group C,high school,free/reduced,none,36,53,43
female,group D,bachelor's degree,free/reduced,none,29,41,47
female,group C,associate's degree,standard,none,62,74,70
female,group C,associate's degree,standard,completed,68,67,73
female,group C,some high school,standard,none,47,54,53
male,group E,associate's degree,standard,completed,62,61,58
female,group E,associate's degree,standard,completed,79,88,94
male,group B,high school,standard,completed,73,69,68
female,group C,bachelor's degree,free/reduced,completed,66,83,83
male,group C,associate's degree,standard,completed,51,60,58
female,group D,high school,standard,none,51,66,62
male,group E,bachelor's degree,standard,completed,85,66,71
male,group A,associate's degree,standard,completed,97,92,86
male,group C,high school,standard,completed,75,69,68
male,group D,associate's degree,free/reduced,completed,79,82,80
female,group C,associate's degree,standard,none,81,77,79
female,group D,associate's degree,standard,none,82,95,89
female,group D,master's degree,standard,none,64,63,66
male,group E,some high school,free/reduced,completed,78,83,80
female,group A,some high school,standard,completed,92,100,97

male,group C,high school,standard,completed,72,67,64
female,group C,high school,free/reduced,none,62,67,64
male,group C,master's degree,standard,none,79,72,69
male,group C,some high school,free/reduced,none,79,76,65
male,group B,bachelor's degree,free/reduced,completed,87,90,88
female,group B,associate's degree,standard,none,40,48,50
male,group D,some college,free/reduced,none,77,62,64
male,group E,associate's degree,standard,none,53,45,40
female,group C,some college,free/reduced,none,32,39,33
female,group C,associate's degree,standard,completed,55,72,79
male,group C,master's degree,free/reduced,none,61,67,66
female,group B,associate's degree,free/reduced,none,53,70,70
male,group D,some high school,standard,none,73,66,62
female,group D,some college,standard,completed,74,75,79
female,group C,some college,standard,none,63,74,74
male,group C,bachelor's degree,standard,completed,96,90,92
female,group D,some college,free/reduced,completed,63,80,80
male,group B,bachelor's degree,free/reduced,none,48,51,46
male,group B,associate's degree,standard,none,48,43,45
female,group E,bachelor's degree,free/reduced,completed,92,100,100
female,group D,master's degree,free/reduced,completed,61,71,78
male,group B,high school,free/reduced,none,63,48,47
male,group D,bachelor's degree,free/reduced,none,68,68,67
male,group B,some college,standard,completed,71,75,70
male,group A,bachelor's degree,standard,none,91,96,92
female,group C,some college,standard,none,53,62,56
female,group C,high school,free/reduced,completed,50,66,64
female,group E,high school,standard,none,74,81,71
male,group A,associate's degree,free/reduced,completed,40,55,53
male,group A,some college,standard,completed,61,51,52
female,group B,high school,standard,none,81,91,89
female,group B,some college,free/reduced,completed,48,56,58
female,group D,master's degree,standard,none,53,61,68
female,group D,some high school,standard,none,81,97,96
female,group E,some high school,standard,none,77,79,80
female,group D,bachelor's degree,free/reduced,none,63,73,78
female,group D,associate's degree,standard,completed,73,75,80
female,group D,some college,standard,none,69,77,77
female,group C,associate's degree,standard,none,65,76,76
female,group A,high school,standard,none,55,73,73
female,group C,bachelor's degree,free/reduced,none,44,63,62
female,group C,some college,standard,none,54,64,65
female,group A,some high school,standard,none,48,66,65
male,group C,some college,free/reduced,none,58,57,54
male,group A,some high school,standard,none,71,62,50
male,group E,bachelor's degree,standard,none,68,68,64
female,group E,high school,standard,none,74,76,73
female,group C,bachelor's degree,standard,completed,92,100,99
female,group C,bachelor's degree,standard,completed,56,79,72
male,group B,high school,free/reduced,none,30,24,15
male,group A,some high school,standard,none,53,54,48
female,group D,high school,standard,none,69,77,73
female,group D,some high school,standard,none,65,82,81
female,group D,master's degree,standard,none,54,60,63
female,group C,high school,standard,none,29,29,30
female,group E,some college,standard,none,76,78,80
male,group D,high school,free/reduced,none,60,57,51
male,group D,master's degree,free/reduced,completed,84,89,90
male,group C,some high school,standard,none,75,72,62
female,group C,associate's degree,standard,none,85,84,82
female,group C,master's degree,free/reduced,none,40,58,54

female,group E,some college,standard,none,61,64,62
female,group B,associate's degree,standard,none,58,63,65
male,group D,some college,free/reduced,completed,69,60,63
female,group C,some college,standard,none,58,59,66
male,group C,bachelor's degree,standard,completed,94,90,91
female,group C,associate's degree,standard,none,65,77,74
female,group A,associate's degree,standard,none,82,93,93
female,group C,high school,standard,none,60,68,72
female,group E,bachelor's degree,standard,none,37,45,38
male,group D,bachelor's degree,standard,none,88,78,83
male,group D,master's degree,standard,none,95,81,84
male,group C,associate's degree,free/reduced,completed,65,73,68
female,group C,high school,free/reduced,none,35,61,54
male,group B,bachelor's degree,free/reduced,none,62,63,56
male,group C,high school,free/reduced,completed,58,51,52
male,group A,some college,standard,completed,100,96,86
female,group E,bachelor's degree,free/reduced,none,61,58,62
male,group D,some college,standard,completed,100,97,99
male,group B,associate's degree,free/reduced,completed,69,70,63
male,group D,associate's degree,standard,none,61,48,46
male,group D,some college,free/reduced,none,49,57,46
female,group C,some high school,standard,completed,44,51,55
male,group D,some college,standard,none,67,64,70
male,group B,high school,standard,none,79,60,65
female,group B,bachelor's degree,standard,completed,66,74,81
female,group C,high school,standard,none,75,88,85
male,group D,some high school,standard,none,84,84,80
male,group A,high school,standard,none,71,74,64
female,group B,high school,free/reduced,completed,67,80,81
female,group D,some high school,standard,completed,80,92,88
male,group E,some college,standard,none,86,76,74
female,group D,associate's degree,standard,none,76,74,73
male,group D,high school,standard,none,41,52,51
female,group D,associate's degree,free/reduced,completed,74,88,90
female,group B,some high school,free/reduced,none,72,81,79
female,group E,high school,standard,completed,74,79,80
male,group B,high school,standard,none,70,65,60
female,group B,bachelor's degree,standard,completed,65,81,81
female,group D,associate's degree,standard,none,59,70,65
female,group E,high school,free/reduced,none,64,62,68
female,group B,high school,standard,none,50,53,55
female,group D,some college,standard,completed,69,79,81
male,group C,some high school,free/reduced,completed,51,56,53
female,group A,high school,standard,completed,68,80,76
female,group D,some college,standard,completed,85,86,98
female,group A,associate's degree,standard,completed,65,70,74
female,group B,some high school,standard,none,73,79,79
female,group B,some college,standard,none,62,67,67
male,group C,associate's degree,free/reduced,none,77,67,64
male,group D,some high school,standard,none,69,66,61
female,group D,associate's degree,free/reduced,none,43,60,58
male,group D,associate's degree,standard,none,90,87,85
male,group C,some college,free/reduced,none,74,77,73
male,group C,some high school,standard,none,73,66,63
female,group D,some college,free/reduced,none,55,71,69
female,group C,high school,standard,none,65,69,67
male,group D,associate's degree,standard,none,80,63,63
female,group C,some high school,free/reduced,completed,50,60,60
female,group C,some college,free/reduced,completed,63,73,71
female,group B,bachelor's degree,free/reduced,none,77,85,87
male,group C,some college,standard,none,73,74,61

male,group D,associate's degree,standard,completed,81,72,77
female,group C,high school,free/reduced,none,66,76,68
male,group D,associate's degree,free/reduced,none,52,57,50
female,group C,some college,standard,none,69,78,76
female,group C,associate's degree,standard,completed,65,84,84
female,group D,high school,standard,completed,69,77,78
female,group B,some college,standard,completed,50,64,66
female,group E,some college,standard,completed,73,78,76
female,group C,some high school,standard,completed,70,82,76
male,group D,associate's degree,free/reduced,none,81,75,78
male,group D,some college,free/reduced,none,63,61,60
female,group D,high school,standard,none,67,72,74
male,group B,high school,standard,none,60,68,60
male,group B,high school,standard,none,62,55,54
female,group C,some high school,free/reduced,completed,29,40,44
male,group B,some college,standard,completed,62,66,68
female,group E,master's degree,standard,completed,94,99,100
male,group E,some college,standard,completed,85,75,68
male,group D,associate's degree,free/reduced,none,77,78,73
male,group A,high school,free/reduced,none,53,58,44
male,group E,some college,free/reduced,none,93,90,83
female,group C,associate's degree,standard,none,49,53,53
female,group E,associate's degree,free/reduced,none,73,76,78
female,group C,bachelor's degree,free/reduced,completed,66,74,81
female,group D,associate's degree,standard,none,77,77,73
female,group C,some high school,standard,none,49,63,56
female,group D,some college,free/reduced,none,79,89,86
female,group C,associate's degree,standard,completed,75,82,90
female,group A,bachelor's degree,standard,none,59,72,70
female,group D,associate's degree,standard,completed,57,78,79
male,group C,high school,free/reduced,none,66,66,59
female,group E,bachelor's degree,standard,completed,79,81,82
female,group B,some high school,standard,none,57,67,72
male,group A,bachelor's degree,standard,completed,87,84,87
female,group D,some college,standard,none,63,64,67
female,group B,some high school,free/reduced,completed,59,63,64
male,group A,bachelor's degree,free/reduced,none,62,72,65
male,group D,high school,standard,none,46,34,36
male,group C,some college,standard,none,66,59,52
male,group D,high school,standard,none,89,87,79
female,group D,associate's degree,free/reduced,completed,42,61,58
male,group C,some college,standard,completed,93,84,90
female,group E,some high school,standard,completed,80,85,85
female,group D,some college,standard,none,98,100,99
male,group D,master's degree,standard,none,81,81,84
female,group B,some high school,standard,completed,60,70,74
female,group B,associate's degree,free/reduced,completed,76,94,87
male,group C,associate's degree,standard,completed,73,78,72
female,group C,associate's degree,standard,completed,96,96,99
female,group C,high school,standard,none,76,76,74
male,group E,associate's degree,free/reduced,completed,91,73,80
female,group C,some college,free/reduced,none,62,72,70
male,group D,some high school,free/reduced,completed,55,59,59
female,group B,some high school,free/reduced,completed,74,90,88
male,group C,high school,standard,none,50,48,42
male,group B,some college,standard,none,47,43,41
male,group E,some college,standard,completed,81,74,71
female,group E,associate's degree,standard,completed,65,75,77
male,group E,some high school,standard,completed,68,51,57
female,group D,high school,free/reduced,none,73,92,84
male,group C,some college,standard,none,53,39,37

female,group B,associate's degree,free/reduced,completed,68,77,80
male,group A,some high school,free/reduced,none,55,46,43
female,group C,some college,standard,completed,87,89,94
male,group D,some high school,standard,none,55,47,44
female,group E,some college,free/reduced,none,53,58,57
male,group C,master's degree,standard,none,67,57,59
male,group C,associate's degree,standard,none,92,79,84
female,group B,some college,free/reduced,completed,53,66,73
male,group D,associate's degree,standard,none,81,71,73
male,group C,high school,free/reduced,none,61,60,55
male,group D,bachelor's degree,standard,none,80,73,72
female,group A,associate's degree,free/reduced,none,37,57,56
female,group C,high school,standard,none,81,84,82
female,group C,associate's degree,standard,completed,59,73,72
male,group B,some college,free/reduced,none,55,55,47
male,group D,associate's degree,standard,none,72,79,74
male,group D,high school,standard,none,69,75,71
male,group C,some college,standard,none,69,64,68
female,group C,bachelor's degree,free/reduced,none,50,60,59
male,group B,some college,standard,completed,87,84,86
male,group D,some high school,standard,completed,71,69,68
male,group E,some college,standard,none,68,72,65
male,group C,master's degree,free/reduced,completed,79,77,75
female,group C,some high school,standard,completed,77,90,85
male,group C,associate's degree,free/reduced,none,58,55,53
female,group E,associate's degree,standard,none,84,95,92
male,group D,some college,standard,none,55,58,52
male,group E,bachelor's degree,free/reduced,completed,70,68,72
female,group D,some college,free/reduced,completed,52,59,65
male,group B,some college,standard,completed,69,77,77
female,group C,high school,free/reduced,none,53,72,64
female,group D,some high school,standard,none,48,58,54
male,group D,some high school,standard,completed,78,81,86
female,group B,high school,standard,none,62,62,63
male,group D,some college,standard,none,60,63,59
female,group B,high school,standard,none,74,72,72
female,group C,high school,standard,completed,58,75,77
male,group B,high school,standard,completed,76,62,60
female,group D,some high school,standard,none,68,71,75
male,group A,some college,free/reduced,none,58,60,57
male,group B,high school,standard,none,52,48,49
male,group D,bachelor's degree,standard,none,75,73,74
female,group B,some high school,free/reduced,completed,52,67,72
female,group C,bachelor's degree,free/reduced,none,62,78,79
male,group B,some college,standard,none,66,65,60
female,group B,some high school,free/reduced,none,49,58,55
female,group B,high school,standard,none,66,72,70
female,group C,some college,free/reduced,none,35,44,43
female,group A,some college,standard,completed,72,79,82
male,group E,associate's degree,standard,completed,94,85,82
female,group D,associate's degree,free/reduced,none,46,56,57
female,group B,master's degree,standard,none,77,90,84
female,group B,high school,free/reduced,completed,76,85,82
female,group C,associate's degree,standard,completed,52,59,62
male,group C,bachelor's degree,standard,completed,91,81,79
female,group B,some high school,standard,completed,32,51,44
female,group E,some high school,free/reduced,none,72,79,77
female,group B,some college,standard,none,19,38,32
male,group C,associate's degree,free/reduced,none,68,65,61
female,group C,master's degree,free/reduced,none,52,65,61
female,group B,high school,standard,none,48,62,60

female,group D,some college,free/reduced,none,60,66,70
male,group D,high school,free/reduced,none,66,74,69
male,group E,some high school,standard,completed,89,84,77
female,group B,high school,standard,none,42,52,51
female,group E,associate's degree,free/reduced,completed,57,68,73
male,group D,high school,standard,none,70,70,70
female,group E,associate's degree,free/reduced,none,70,84,81
male,group E,some college,standard,none,69,60,54
female,group C,associate's degree,standard,none,52,55,57
male,group C,some high school,standard,completed,67,73,68
male,group C,some high school,standard,completed,76,80,73
female,group E,associate's degree,standard,none,87,94,95
female,group B,some college,standard,none,82,85,87
female,group C,some college,standard,none,73,76,78
male,group A,some college,free/reduced,none,75,81,74
female,group D,some college,free/reduced,none,64,74,75
female,group E,high school,free/reduced,none,41,45,40
male,group C,high school,standard,none,90,75,69
male,group B,bachelor's degree,standard,none,59,54,51
male,group A,some high school,standard,none,51,31,36
male,group A,high school,free/reduced,none,45,47,49
female,group C,master's degree,standard,completed,54,64,67
male,group E,some high school,standard,completed,87,84,76
female,group C,high school,standard,none,72,80,83
male,group B,some high school,standard,completed,94,86,87
female,group A,bachelor's degree,standard,none,45,59,64
male,group D,bachelor's degree,free/reduced,completed,61,70,76
female,group B,high school,free/reduced,none,60,72,68
female,group C,some high school,standard,none,77,91,88
female,group A,some high school,standard,completed,85,90,92
female,group D,bachelor's degree,free/reduced,none,78,90,93
male,group E,some college,free/reduced,completed,49,52,51
female,group B,high school,free/reduced,none,71,87,82
female,group C,some high school,free/reduced,none,48,58,52
male,group C,high school,standard,none,62,67,58
female,group C,associate's degree,free/reduced,completed,56,68,70
female,group C,some high school,standard,none,65,69,76
female,group D,some high school,free/reduced,completed,69,86,81
male,group B,some high school,standard,none,68,54,53
female,group A,some college,free/reduced,none,61,60,57
female,group C,bachelor's degree,free/reduced,completed,74,86,89
male,group A,bachelor's degree,standard,none,64,60,58
female,group B,high school,standard,completed,77,82,89
male,group B,some college,standard,none,58,50,45
female,group C,high school,standard,completed,60,64,74
male,group E,high school,standard,none,73,64,57
female,group A,high school,standard,completed,75,82,79
male,group B,associate's degree,free/reduced,completed,58,57,53
female,group C,associate's degree,standard,none,66,77,73
female,group D,high school,free/reduced,none,39,52,46
male,group C,some high school,standard,none,64,58,51
female,group B,high school,free/reduced,completed,23,44,36
male,group B,some college,free/reduced,completed,74,77,76
female,group D,some high school,free/reduced,completed,40,65,64
male,group E,master's degree,standard,none,90,85,84
male,group C,master's degree,standard,completed,91,85,85
male,group D,high school,standard,none,64,54,50
female,group C,high school,standard,none,59,72,68
male,group D,associate's degree,standard,none,80,75,69
male,group C,master's degree,standard,none,71,67,67
female,group A,high school,standard,none,61,68,63

female,group E,some college,standard,none,87,85,93
male,group E,some high school,standard,none,82,67,61
male,group C,some high school,standard,none,62,64,55
female,group B,bachelor's degree,standard,none,97,97,96
male,group B,some college,free/reduced,none,75,68,65
female,group C,bachelor's degree,standard,none,65,79,81
male,group B,high school,standard,completed,52,49,46
male,group C,associate's degree,free/reduced,none,87,73,72
female,group C,associate's degree,standard,none,53,62,53
female,group E,master's degree,free/reduced,none,81,86,87
male,group D,bachelor's degree,free/reduced,completed,39,42,38
female,group C,some college,standard,completed,71,71,80
male,group C,associate's degree,standard,none,97,93,91
male,group D,some college,standard,completed,82,82,88
male,group C,high school,free/reduced,none,59,53,52
male,group B,associate's degree,standard,none,61,42,41
male,group E,associate's degree,free/reduced,completed,78,74,72
male,group C,associate's degree,free/reduced,none,49,51,51
male,group B,high school,standard,none,59,58,47
female,group C,some college,standard,completed,70,72,76
male,group B,associate's degree,standard,completed,82,84,78
male,group E,associate's degree,free/reduced,none,90,90,82
female,group C,bachelor's degree,free/reduced,none,43,62,61
male,group C,some college,free/reduced,none,80,64,66
male,group D,some college,standard,none,81,82,84
male,group C,some high school,standard,none,57,61,54
female,group D,some high school,standard,none,59,72,80
female,group D,associate's degree,standard,none,64,76,74
male,group C,bachelor's degree,standard,completed,63,64,66
female,group E,bachelor's degree,standard,completed,71,70,70
female,group B,high school,free/reduced,none,64,73,71
male,group D,bachelor's degree,free/reduced,none,55,46,44
female,group E,associate's degree,standard,none,51,51,54
female,group C,associate's degree,standard,completed,62,76,80
female,group E,associate's degree,standard,completed,93,100,95
male,group C,high school,free/reduced,none,54,72,59
female,group D,some college,free/reduced,none,69,65,74
male,group D,high school,free/reduced,none,44,51,48
female,group E,some college,standard,completed,86,85,91
female,group E,associate's degree,standard,none,85,92,85
female,group A,master's degree,free/reduced,none,50,67,73
male,group D,some high school,standard,completed,88,74,75
female,group E,associate's degree,standard,none,59,62,69
female,group E,some high school,free/reduced,none,32,34,38
male,group B,high school,free/reduced,none,36,29,27
female,group B,some high school,free/reduced,completed,63,78,79
male,group D,associate's degree,standard,completed,67,54,63
female,group D,some high school,standard,completed,65,78,82
male,group D,master's degree,standard,none,85,84,89
female,group C,master's degree,standard,none,73,78,74
female,group A,high school,free/reduced,completed,34,48,41
female,group D,bachelor's degree,free/reduced,completed,93,100,100
female,group D,some high school,free/reduced,none,67,84,84
male,group D,some college,standard,none,88,77,77
male,group B,high school,standard,none,57,48,51
female,group D,some college,standard,completed,79,84,91
female,group C,bachelor's degree,free/reduced,none,67,75,72
male,group E,bachelor's degree,standard,completed,70,64,70
male,group D,bachelor's degree,free/reduced,none,50,42,48
female,group A,some college,standard,none,69,84,82
female,group C,bachelor's degree,standard,completed,52,61,66

female,group C,bachelor's degree,free/reduced,completed,47,62,66
female,group B,associate's degree,free/reduced,none,46,61,55
female,group E,some college,standard,none,68,70,66
male,group E,bachelor's degree,standard,completed,100,100,100
female,group C,high school,standard,none,44,61,52
female,group C,associate's degree,standard,completed,57,77,80
male,group B,some college,standard,completed,91,96,91
male,group D,high school,free/reduced,none,69,70,67
female,group C,high school,free/reduced,none,35,53,46
male,group D,high school,standard,none,72,66,66
female,group B,associate's degree,free/reduced,none,54,65,65
male,group D,high school,free/reduced,none,74,70,69
male,group E,some high school,standard,completed,74,64,60
male,group E,associate's degree,free/reduced,none,64,56,52
female,group D,high school,free/reduced,completed,65,61,71
male,group E,associate's degree,free/reduced,completed,46,43,44
female,group C,some high school,free/reduced,none,48,56,51
male,group C,some college,free/reduced,completed,67,74,70
male,group D,some college,free/reduced,none,62,57,62
male,group D,associate's degree,free/reduced,completed,61,71,73
male,group C,bachelor's degree,free/reduced,completed,70,75,74
male,group C,associate's degree,standard,completed,98,87,90
male,group D,some college,free/reduced,none,70,63,58
male,group A,associate's degree,standard,none,67,57,53
female,group E,high school,free/reduced,none,57,58,57
male,group D,some college,standard,completed,85,81,85
male,group D,some high school,standard,completed,77,68,69
male,group C,master's degree,free/reduced,completed,72,66,72
female,group D,master's degree,standard,none,78,91,96
male,group C,high school,standard,none,81,66,64
male,group A,some high school,free/reduced,completed,61,62,61
female,group B,high school,standard,none,58,68,61
female,group C,associate's degree,standard,none,54,61,58
male,group B,high school,standard,none,82,82,80
female,group D,some college,free/reduced,none,49,58,60
male,group B,some high school,free/reduced,completed,49,50,52
female,group E,high school,free/reduced,completed,57,75,73
male,group E,high school,standard,none,94,73,71
female,group D,some college,standard,completed,75,77,83
female,group E,some high school,free/reduced,none,74,74,72
male,group C,high school,standard,completed,58,52,54
female,group C,some college,standard,none,62,69,69
male,group E,associate's degree,standard,none,72,57,62
male,group C,some college,standard,none,84,87,81
female,group D,master's degree,standard,none,92,100,100
female,group D,high school,standard,none,45,63,59
male,group C,high school,standard,none,75,81,71
female,group A,some college,standard,none,56,58,64
female,group D,some high school,free/reduced,none,48,54,53
female,group E,associate's degree,standard,none,100,100,100
female,group C,some high school,free/reduced,completed,65,76,75
male,group D,some college,standard,none,72,57,58
female,group D,some college,standard,none,62,70,72
male,group A,some high school,standard,completed,66,68,64
male,group C,some college,standard,none,63,63,60
female,group E,associate's degree,standard,none,68,76,67
female,group B,bachelor's degree,standard,none,75,84,80
female,group D,bachelor's degree,standard,none,89,100,100
male,group C,some high school,standard,completed,78,72,69
female,group A,high school,free/reduced,completed,53,50,60
female,group D,some college,free/reduced,none,49,65,61

female,group A,some college,standard,none,54,63,67
female,group C,some college,standard,completed,64,82,77
male,group B,some college,free/reduced,completed,60,62,60
male,group C,associate's degree,standard,none,62,65,58
male,group D,high school,standard,completed,55,41,48
female,group C,associate's degree,standard,none,91,95,94
female,group B,high school,free/reduced,none,8,24,23
male,group D,some high school,standard,none,81,78,78
male,group B,some high school,standard,completed,79,85,86
female,group A,some college,standard,completed,78,87,91
female,group C,some high school,standard,none,74,75,82
male,group A,high school,standard,none,57,51,54
female,group C,associate's degree,standard,none,40,59,51
male,group E,some high school,standard,completed,81,75,76
female,group A,some high school,free/reduced,none,44,45,45
female,group D,some college,free/reduced,completed,67,86,83
male,group E,high school,free/reduced,completed,86,81,75
female,group B,some high school,standard,completed,65,82,78
female,group D,associate's degree,free/reduced,none,55,76,76
female,group D,bachelor's degree,free/reduced,none,62,72,74
male,group A,high school,standard,none,63,63,62
female,group E,master's degree,standard,completed,88,99,95
male,group C,high school,free/reduced,none,62,55,55
female,group C,high school,free/reduced,completed,59,71,65
female,group D,some college,standard,completed,68,78,77
female,group D,some college,free/reduced,none,77,86,86

In [47]:

```
import pandas as pd
df = pd.read_csv('G:\\Python Session\\archive\\StudentsPerformance.csv')
print(df)
```

	gender	race/ethnicity	parental level of education	lunch	
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	
..	...	...	...	...	
995	female	group E	master's degree	standard	
996	male	group C	high school	free/reduced	
997	female	group C	high school	free/reduced	
998	female	group D	some college	standard	
999	female	group D	some college	free/reduced	

	test preparation course	math score	reading score	writing score
0	none	72	72	74
1	completed	69	90	88
2	none	90	95	93
3	none	47	57	44
4	none	76	78	75
..	...	...	...	...
995	completed	88	99	95
996	none	62	55	55
997	completed	59	71	65
998	completed	68	78	77
999	none	77	86	86

[1000 rows x 8 columns]

In [50]:

df[0:5]

Out[50]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

In [56]:

```
df[995:1001]
```

Out[56]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
995	female	group E	master's degree	standard	completed	88	99	95
996	male	group C	high school	free/reduced	none	62	55	55
997	female	group C	high school	free/reduced	completed	59	71	65
998	female	group D	some college	standard	completed	68	78	77
999	female	group D	some college	free/reduced	none	77	86	86

In []:

```
#List of male student having math score > 80
```